

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2024

Course Code: **PHY101R01**

Course: **ENGINEERING PHYSICS**

QP No. :U178-2

Duration: 3 hours

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Obtain the expression to find the wavelength of an electron accelerated by a high potential difference.
2. State Heisenberg Uncertainty principle.
3. Sketch the variation of density of states in 3D, 2D, 1D and 0D materials.
4. For a laser diode, the wavelength of emitted light is $1.55\mu\text{m}$. Determine its band gap in eV. 2.28×10^{-19}
5. An optical fiber has core fiber index of 1.5 and relative difference in refractive index $\Delta=0.015$. Determine the numerical aperture and the acceptance angle of the fiber. 0.23
6. Calculate the electronic polarizability of Argon atom with a relative permittivity of 1.0024 at NTP with 2.7×10^{25} atoms/ m^3 . 3.7128×10^{-40}
7. List any four energies associated with Domain theory of ferromagnetism.

8. What is Curie temperature in Ferromagnetic material? Give the expression for magnetic susceptibility based on Curie temperature.
9. State law of mass action.
10. At 20°C, the conductivity of Ge is $2 \Omega^{-1}\text{m}^{-1}$. Find the conductivity of Ge at 40 °C. ($E_g = 0.72 \text{ eV}$).

PART – B

Answer any FOUR questions

4 x 15 = 60 Marks

11. An electron is trapped in a box. Derive the expression for energy of an electron and obtain the wave function. Also discuss its results.
12. Explain the construction and working of TEA laser with appropriate energy level diagrams.
13. Classify the Fibers based on the refractive index profile and mode of communication. From your understandings, suggest a suitable fiber for communication purpose.
14. Compare different types of magnetic materials based on its properties.
15. Derive an equation to find the local field. Also, derive an equation which relates macroscopic dielectric constant with microscopic internal field.
16. How will you estimate the concentration of holes in the valence band of p-type semiconductor? Derive the appropriate equation.

PART – C

Answer the following

1 x 20 = 20 Marks

17. a) A metal surface with a work function of 2.2 eV is irradiated with a wavelength of 300 nm. Calculate its kinetic energy of photoelectron. (3)

$$2.2 \text{ eV}$$

$$- 2.37 \times 10^{-19} \text{ J}$$

$$2$$

- b) Determine Numerical aperture and V-number of a fiber. Given $\lambda = 1000 \text{ nm}$, $n_{\text{core}} = 1.46$, $n_{\text{cladding}} = 1.45$ and core radius is $5 \mu\text{m}$. (3)
- c) Based on the domain theory of ferromagnetism explain the B-H curve. (6)
- d) Explain an experimental procedure to determine the charge carrier density of an extrinsic semiconducting material. (8)
